INCH-POUND

The documentation and process conversion measures necessary to comply with this document shall be completed by 5 September 2021.

MIL-PRF-19500/446F
w/AMENDMENT 1
5 May 2021
SUPERSEDING
MIL-PRF-19500/446F
26 June 2019

PERFORMANCE SPECIFICATION SHEET

RECTIFIER, SEMICONDUCTOR DEVICE, UNITIZED, SILICON, HIGH-POWER, SINGLE PHASE, FULL WAVE BRIDGE RECTIFIER, TYPES SPA25, SPB25, SPC25, AND SPD25, QUALITY LEVELS JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, single phase, full wave bridge rectifiers. Two levels of product assurance (JANTX and JANTXV) are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

1.3.1 Ratings applicable to all Part or Identifying Numbers (PIN). Operating temperature: −65°C to +150°C. Storage ambient temperature: −65°C to +150°C.

1.3.2 Ratings applicable to individual types.

Type	V_{RWM}	Bridge (1) I_{O1} at T_C = +55°C	Bridge (2) I_{O2} at T_C = +100°C	$I_{F(surge)}$ I_O = 25 A dc T_C = +55°C t_p = 8.3 ms	Barometric pressure reduced	t_{rr} I_F = 0.5 A I_R = 1.0 A I_{rec} = 0.25 A
	V (pk)	A dc	A dc	A (pk)	mm Hg	µs
SPA25	100	25	15	150	8	2.5
SPB25	200	25	15	150	8	2.5
SPC25	400	25	15	150	8	2.5
SPD25	600	25	15	150	8	2.5

(1) Derate from 25 A dc at T_C = +55°C to 15 A dc at +100°C (222 mA dc/°C).
(2) Derate from 15 A dc at T_C = +100°C to 0 A dc at +150°C (300 mA dc/°C).

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1.5 PIN. The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. The JAN certification mark and quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANTX" and "JANTXV".

1.5.2 Type designations. The identification system for this specification sheet uses the following type designations: "SPA25", "SPB25", "SPC25", and "SPD25".

1.5.3 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods For Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

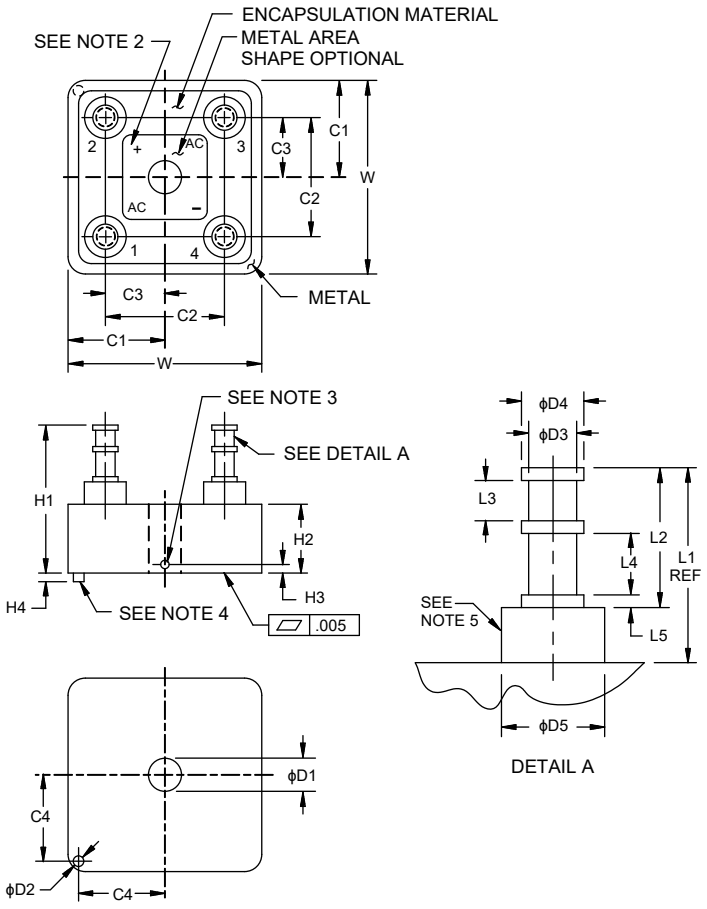
3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
C1	.552	.572	14.02	14.53
C2	.624	.760	15.85	19.30
C3	.312	.380	7.92	9.65
C4	.495	.512	12.57	13.00
ϕD1	.189	.195	4.80	4.95
ϕD2	.057	.067	1.45	1.70
ϕD3	.108	.118	2.74	3.00
ϕD4	.141	.151	3.58	3.84
ϕD5	.225	.260	5.72	6.60
H1	.690	1.060	17.53	26.92
H2	.300	.500	7.62	12.70
H3	.040	.060	1.02	1.52
H4	.042	.062	1.07	1.57
L1	.370	.560	9.40	14.22
L2	.307	.365	7.80	9.27
L3	.059	.099	1.49	2.52
L4	.132	.163	3.35	4.14
L5	.026	.036	0.66	0.91
W	1.104	1.144	28.04	29.06

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Polarity shall be marked on the terminal side of the device. Terminal numbers are for reference and do not have to be marked on the bridge.
3. Point at which T_C is read shall be in metal part of case as shown on figure.
4. The locating pin shall be adjacent to the positive (+) terminal.
5. Insulating sleeve shall be alumina (Al_2O_3) composite material or equivalent.
6. The areas defined by the dimensions "L3" and "L4" are both acceptable for wire wrap and soldering.
7. In accordance with ASME Y14.5, diameters are equivalent to Øx symbology.

FIGURE 1. Physical dimensions for bridge rectifier assembly.

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3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Internal construction. The bridge rectifier shall consist of a metal and plastic encased assembly of discrete diodes. Multiple diodes for each leg construction shall not be permitted.

3.4.1.1 Discrete diodes. Each discrete diode shall be a glass-to-metal, ceramic-to-metal, or fused metal oxide-to-metal hermetically sealed package. The silicon die in each discrete diode shall be metallurgically bonded directly to the terminal pins. Discrete diodes shall be JANTX or JANTXV level devices, or equivalent screened non-JAN devices manufactured in a MIL-PRF-19500 certified facility. Only those discrete diodes which have met these requirements shall be used in the bridge rectifier.

3.4.1.2 Completed assemblies. The completed assembly of diodes and other internal structures shall be encapsulated in a plastic material which polymerizes to a rigid condition by virtue of a chemical cross-linking mechanism. The bridge rectifier shall be free of voids either visible or as evidenced by failure to pass the environment test specified herein.

3.4.2 Terminal finish. Terminal finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of terminal finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. Manufacturer's identification and date code shall be marked on the devices. Initial container package marking shall be in accordance with MIL-PRF-19500. Polarity shall be as marked on figure 1.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.3 Screening (JANTX and JANTXV). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable. The JANTXV requirements apply to the internal discrete diodes only, not to the assembly.

4.3.1 Discrete diode screening. One hundred percent of the internal discrete diodes shall be subjected to the following inspections.

Screen	Measurement
3c (1)	Thermal impedance (see 4.5.3)
9	Not applicable
11	I_{R1} and V_{F1}
12	See 4.3.1.1
13	Subgroup 2 of table I herein. ΔV_{F1} = ±0.1 V dc; ΔI_{R1} = 100 percent of initial value or ±250 nA dc, whichever is greater

(1) This test shall be performed any time after temperature cycling, screen 3a, and does not need to be repeated in screening requirements.

4.3.1.1 Power burn-in conditions. The power burn-in conditions shall be in accordance with test condition B of method 1038 of MIL-STD-750. The following details shall apply: T_A = room ambient as defined in the general requirements of MIL-STD-750 or 4.5 herein, V_{RWM} = rated V_{RWM} (see 1.3), I_O = 3 A dc, f = 50 to 60 Hz.

4.3.2 Assembly screening. One hundred percent of the completed bridge rectifier assemblies shall be subjected to the following inspections.

Screen	Measurement
Temperature cycling	Method 1051 of MIL-STD-750, condition F.
End point electrical measurements (1)	Subgroup 2 of table I herein.
Dielectric withstanding voltage	See 4.5.1.

(1) Except thermal impedance.

4.4 Conformance inspection. Conformance inspection shall be performed on the finished bridge rectifier assemblies in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JANTX and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 (except thermal impedance) herein.

Subgroup	Method	Conditions
B1	2026	Immerse to insulating sleeve.
B2	1051	Condition F, 25 cycles.
B2	1071	Not applicable.
B3	1027	I_O = 0 A dc; T_A = 150°C; V_{RWM} = 80 percent rated V_{RWM} (see 1.3), f = 50 to 60 Hz, n = 10, c = 0.
B5		Operational thermal cycling (see 4.5.4).
B5	3105	Junction temperature test (see 4.5.2).
B6		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Group C sample size may be modified, when applicable, per the subgroups below.

Subgroup	Method	Conditions
C2	1056	Not applicable.
C2	2036	Test condition A, 6 pounds (2.72 kg), t = 30 seconds.
C2	2036	Test condition D1, t = 10 seconds, 5 inch-pounds (0.56 N-m).
C2	1071	Not applicable.
C6	1026	I_O = 0 A dc; T_A = 150°C; V_{RWM} = 80 percent rated V_{RWM} (see 1.3), f = 50 to 60 Hz, sampling plan n = 10, c = 0.
C8	1001	Pressure = 8 mm Hg; I_R = 2.0 µA dc maximum, V_R = rated V_{RWM} (see 1.3), (applied between all terminals to the case), t = 60 seconds; sampling plan n = 12, c = 0.
C8	4066	V_{RWM} = 0 V, bridge I_O = 25 A dc (entire bridge biased); T_C = 55°C + 10°C, −0°C; I_{FSM} = 150 A (pk) (each device for each bridge leg); tp = 8.3 ms; ten surges for each leg at maximum 1 minute intervals; sampling plan n = 22, c = 0. Alternate condition: Each leg may be biased separately to I_O = 12.5 A dc. All other conditions are the same as above.

4.4.3.1 Sampling. The following allowances for small lot samples (samples from small inspection lots) shall only apply when the discrete diodes are screened to quality level JANTX or JANTXV requirements (see 6.7).

a. Small lot samples may be used to satisfy the group C requirements for inspection lots of structurally identical devices.

b. Small lot samples may be used to satisfy the 1-year group C interval as allowed for large lot samples in MIL-PRF-19500.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Dielectric withstanding voltage. This test shall be performed in accordance with method 1081 of MIL–STD–750. The metal case of the rectifier assembly shall be connected to ground and all terminals connected to the high potential side of a dc power supply or a scope display test set. The voltage applied between the terminals and the case, shall be 2,800 volts dc and shall be held at that level for 10 seconds. Any discontinuity or dynamic instability of the trace, or a breakdown current in excess of 10 µA dc, shall be cause for rejection.

4.5.2 Junction temperature test. The junction temperature test shall be performed in accordance with method 3105 of MIL-STD-750. The maximum junction temperature for any diode in the bridge rectifier at the below specified conditions shall be 175°C maximum. The test conditions shall be one of the following:

a. T_C = 55°C, I_O = 25 A dc, I_{ref} = 10 mA dc or

b. T_C = 100°C, I_O = 15 A dc, I_{ref} = 10 mA dc.

4.5.3 Thermal impedance. The thermal impedance measurement shall be performed in accordance with method 3101 of MIL-STD-750. The following parameters shall apply:

a. Measurement current (I_M): 10 mA;

b. Heating current (I_H): 10 A minimum;

c. Heating time (t_H): 10 ms;

d. Measurement time delay (t_{MD}): 70 µs maximum.

The maximum limit for $Z_{\theta JX}$ under these test conditions is 1.5°C/W.

4.5.4 Operational thermal cycling. This test shall be performed by connecting a single phase 50 to 60 Hz sine wave input to the ac terminals of the bridge rectifier. The resistive load or input voltage shall be made variable in order to maintain a rated bridge output current of 25 A dc. The bridge rectifier case temperature shall be maintained at 55°C except during the cooling cycle when case temperature shall be permitted to drop 20°C maximum. Life-test duration shall be 2,000 cycles with a heating time of 8 minutes, +10, −0 minutes and a cooling time ≥ 2 minutes.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3101	See 4.5.3	$Z_{\theta JX}$		1.5	°C/W
Forward voltage 2/	4011	Condition B; t_p = 8.3 ms maximum; duty cycle ≤ 2 percent; I_F = 39 A dc (pk),	V_{F1}	0.9	1.40	V (pk)
Reverse current 2/	4016	Condition A or B; V_R = rated V_{RWM} (see 1.3)	I_{R1}		2.0	µA dc
Breakdown voltage 2/	4021	I_R = 50 µA dc	$V_{(BR)}$			
SPA25				110		V dc
SPB25				220		V dc
SPC25				440		V dc
SPD25				660		V dc
Subgroup 3						
High temperature operation:		T_C = 100°C				
Reverse current 2/	4016	Condition A or B; V_R = rated V_{RWM} (see 1.3)	I_{R2}		150	µA dc
Low temperature operation:		T_A = −55°C				
Reverse current 2/	4016	Condition A or B; V_R = rated V_{RWM} (see 1.3)	I_{R3}		50	µA dc
Subgroup 4						
Dielectric withstanding voltage	1081	See 4.5.1				
Reverse recovery time 2/	4031	Condition B1; I_F = 0.5 A, I_R = 1.0 A, I_{rec} = 0.25 A	t_{rr}		2.5	µs
Subgroups 5 and 6						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.
2/ Applies to each leg of the bridge rectifier.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

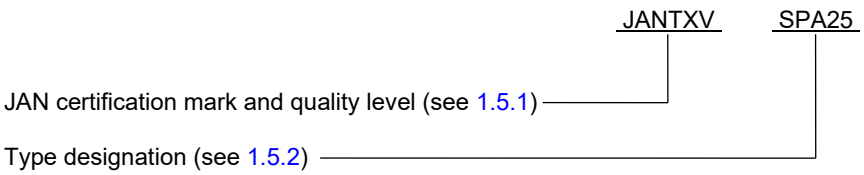
c. Terminal finish (see 3.4.2).

d. The complete PIN, see 1.5 and 6.4.

e. Destructive physical analysis when requested.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil . An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for the device assemblies are constructed using the following form.



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6.5 List of PINs. The following is a list of all possible PINs for the devices available on this specification sheet.

Quality level TX	JANTXSPA25	JANTXSPB25	JANTXSPC25	JANTXSPD25
Quality level TXV	JANTXVSPA25	JANTXVSPB25	JANTXVSPC25	JANTXVSPD25

6.6 Interchangeability. Quality level JAN bridge rectifiers manufactured to comply with MIL-PRF-19500/446D and previous revisions were constructed with JANTX level discrete diodes and are the same product assurance level as the JANTX bridge rectifiers in this revision. Quality level JANTX bridge rectifiers are identical for all requirements of this specification except for the device marking and are directly interchangeable for quality level JAN bridge rectifiers. The JANTX bridge rectifier is the preferred PIN for new design and acquisition. Quality level JAN bridge rectifiers are no longer offered under this specification as of MIL-PRF-19500/446E dated 15 December 2008. Existing stock of quality level JAN bridge rectifiers may be used until exhausted.

6.7 Group C sampling modification. The use of samples from small inspection lots (small lot samples) for group C inspection is allowed in this specification sheet because of the unique design and construction of the end item rectifier bridge that consists of multiple screened non-cavity discrete diodes which are assembled and then encapsulated.

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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